Satellite Data for Agricultural and Soil-Health Evaluation

A detailed account of how data are obtained, processed, modelled and evaluated | October 2026

Part A. Physical basis: what a satellite actually measures

A satellite sensor records electromagnetic energy leaving the Earth's surface. For agriculture the useful regions are: **visible (400-700 nm)**, where chlorophyll absorbs strongly in blue and red; the **red edge (680-750 nm)**, where reflectance rises steeply and whose position shifts with chlorophyll content and nitrogen status; the **near-infrared (750-1300 nm)**, where healthy leaf mesophyll scatters strongly so that canopy reflectance increases with leaf area; the **shortwave infrared (1300-2500 nm)**, dominated by water absorption in vegetation and by clay-mineral (~2200 nm), carbonate (~2340 nm) and organic-matter features in soils; the **thermal infrared (8-14 µm)**, which gives land-surface temperature and hence evaporative cooling and crop water stress; and the **microwave (1-30 cm)**, where active radar backscatter depends on surface roughness, canopy structure and dielectric constant (i.e. water content), and passive radiometers sense near-surface soil moisture.

Soil reflectance in the 400-2500 nm range decreases overall as organic carbon increases (soils darken), shows iron-oxide features near 500-900 nm, hydroxyl/clay features near 1400, 1900 and 2200 nm, and carbonate features near 2340 nm. These are the physical hooks that allow SOC, clay, $CaCO_3$ and iron to be estimated from spectra. pH, CEC, total N and microbial properties lack distinct features and are predicted through covariance with the directly sensed constituents and with landscape covariates.

Part B. Obtaining the data

B.1 Sensor families and representative missions

Sensor family	Representative missions (free data)	Spatial / temporal resolution	Agricultural use
Optical multispectral	Sentinel-2A/B/C (ESA); Landsat 8/9 (USGS/NASA); MODIS Terra/Aqua, VIIRS (NASA/NOAA)	S2: 10/20/60 m, ~5 d; Landsat: 30 m (15 m pan, 100 m thermal), 8 d combined; MODIS: 250-1000 m, daily	Vegetation indices, LAI/fAPAR, crop type, phenology, bare-soil spectra
Hyperspectral (imaging spectroscopy)	EnMAP (DLR), PRISMA (ASI), EMIT (NASA, on ISS), DESIS (DLR, on ISS)	30 m, 200+ bands; tasking-based or limited coverage	Soil mineralogy and SOC, pigment and nitrogen retrieval, species / trait mapping
Synthetic aperture radar (SAR)	Sentinel-1A/C (C-band, ESA); NISAR (L/S-band, NASA-ISRO); ALOS-2 (L-band, JAXA)	5-20 m, 6-12 d; cloud-independent, day and night	Crop structure and type, flooding, soil moisture, tillage, phenology under persistent cloud
Thermal infrared	Landsat TIRS; ECOSTRESS (ISS); MODIS/VIIRS; Sentinel-3 SLSTR	70-1000 m	Land-surface temperature, evapotranspiration, water-stress indices
Passive microwave	SMAP (NASA); SMOS (ESA); AMSR2 (JAXA)	9-40 km, 1-3 d	Regional surface soil moisture, drought monitoring, crop-model forcing
Commercial very-high-resolution	PlanetScope (3 m, daily); SkySat; WorldView; Pleiades	0.3-3 m, daily to on-demand	Small-holder fields, within-field zoning, emergence counts (not free; some research programmes provide access)

B.2 Processing levels and what to download

Agencies distribute data at standard levels. **Level-1** products are radiometrically calibrated and geo-located top-of-atmosphere (TOA) radiance or reflectance (e.g. Sentinel-2 L1C, Landsat L1TP). **Level-2** products have been atmospherically corrected to *surface reflectance* and include per-pixel quality masks (e.g. Sentinel-2 L2A with its Scene Classification Layer, Landsat Collection 2 Level-2 with the QA_PIXEL band, Sentinel-1 GRD after terrain correction). **Level-3/4** products are gridded, composited or modelled variables (MODIS 16-day NDVI, Copernicus Global Land LAI/fAPAR, SMAP soil moisture, HLS). For most agricultural analysis one starts from Level-2 surface reflectance; analysis-ready data (ARD) collections on cloud platforms remove most of the remaining preparation burden.

B.3 Access routes

- **Agency portals** (Copernicus Data Space Ecosystem, USGS EarthExplorer, NASA Earthdata) for scene download and APIs (STAC, OData, openEO).
- **Cloud platforms** that co-locate data and compute: Google Earth Engine (multi-petabyte catalogue, JavaScript/Python API; Gorelick et al., 2017), Microsoft Planetary Computer, AWS Open Data registry, Copernicus Data Space JupyterHub / Sentinel Hub.
- **Thematic services** that deliver finished variables: Copernicus Land Monitoring Service, FAO WaPOR, GEOGLAM Crop Monitor, USDA NASS CropScape.

Part C. Pre-processing to analysis-ready data

C.1 Radiometric calibration

Raw digital numbers are converted to at-sensor radiance using gain/offset coefficients, then to TOA reflectance by normalising for solar irradiance, Earth-Sun distance and solar zenith angle. Modern missions are cross-calibrated against each other and against vicarious targets so that time series are radiometrically stable to within ~1-3%.

C.2 Atmospheric correction

Scattering and absorption by aerosols, water vapour and ozone are removed to obtain surface reflectance. Operational processors include Sen2Cor (Sentinel-2 L2A), LaSRC and LEDAPS (Landsat), MAJA (multi-temporal, Theia/CNES), FORCE, and physics-based codes such as 6S and MODTRAN. Over dark soils and sparse canopies, residual aerosol errors can bias soil-property models, so consistent processor choice across dates is important.

C.3 Cloud, shadow and snow screening

Optical imagery is unusable under cloud. Masks come from the product itself (Sentinel-2 SCL and Cloud Probability, Landsat Fmask/QA_PIXEL), from machine-learning detectors (s2cloudless), or from multi-temporal consistency tests. Aggressive masking plus temporal compositing (median, maximum-NDVI or best-available-pixel) yields cloud-free mosaics for a target window. In cloudy regions Sentinel-1 SAR time series are fused to fill gaps.

C.4 Geometric processing and harmonisation

Orthorectification with a digital elevation model places pixels on a map grid (UTM/WGS84 tiles for Sentinel-2 and Landsat). Multi-sensor work requires band-pass adjustment and BRDF normalisation; NASA's Harmonized Landsat Sentinel-2 (HLS) product does this to produce a 30 m, 2-3 day virtual constellation. SAR data require calibration to sigma-0 or gamma-0, speckle filtering and radiometric terrain correction.

C.5 Time-series preparation

Irregularly spaced, gappy observations are regularised by interpolation and smoothing (Savitzky-Golay, Whittaker, harmonic / Fourier fitting, double-logistic phenology models). Outputs include per-pixel phenological metrics (start, peak and end of season, length, amplitude, integrated greenness) that are strong predictors of crop type and yield.

C.6 Bare-soil compositing for soil properties

Topsoil can only be sensed where it is exposed. Bare-soil composites select, over several years, every pixel-date where NDVI is low (commonly < 0.25-0.30), a bare-soil or normalised burn index indicates no residue or dry vegetation, and no cloud/shadow is present, then aggregate the surface reflectance (median or mean) per pixel. Examples are the Soil Composite Mapping Processor (SCMaP) for Landsat and the Geospatial Soil Sensing System (GEOS3). Further filters remove moist soils (SWIR-based moisture indices) and rough, freshly tilled surfaces, both of which distort spectra. The composite is then matched to soil-sample locations for model training.

Part D. Feature derivation

Feature	Formula / method	Sensitive to	Comments
NDVI	(NIR - Red) / (NIR + Red)	Green biomass, fractional cover	Saturates at high LAI; the workhorse index (Tucker, 1979)
EVI	2.5 (NIR - Red) / (NIR + 6 Red - 7.5 Blue + 1)	Biomass, with reduced saturation and atmospheric sensitivity	MODIS standard product
NDRE / CIred-edge	(NIR - RE) / (NIR + RE); NIR/RE - 1	Canopy chlorophyll, nitrogen	Requires red-edge bands (Sentinel-2 B5-B7)
SAVI / MSAVI	(1 + L)(NIR - Red) / (NIR + Red + L)	Vegetation with soil-brightness correction	Useful at early growth stages
NDMI / NDWI	(NIR - SWIR1) / (NIR + SWIR1)	Canopy and soil water content	Also used to filter moist bare soil
Bare-soil indices	BSI, NBR2 = (SWIR1 - SWIR2)/(SWIR1 + SWIR2)	Soil exposure, residue, moisture	Thresholds select pixels for soil composites
Soil spectral features	Band reflectances, ratios, continuum removal, derivatives, PCA	SOC, clay, carbonates, iron	Basis for PLSR / Cubist soil models; S2 B11/B12 and visible bands most informative
LAI, fAPAR, CCC	PROSAIL inversion (look-up table or neural network) or empirical regression	Canopy structure and pigments	ESA SNAP Biophysical Processor; Copernicus Global Land products
SAR features	VV, VH backscatter, VH/VV ratio, coherence, polarimetric decompositions	Canopy structure, moisture, roughness	Cloud-independent; strong for rice, flooding, tillage
Thermal features	Land-surface temperature, Ts - Ta, evaporative fraction	Water stress, ET	Energy-balance models (SEBAL, METRIC, TSEB)

Part E. Modelling approaches

E.1 Empirical / statistical

Linear and multiple regression, partial least squares regression (PLSR) and geostatistical models (kriging, regression-kriging, linear spatial mixed models) remain widely used because they are interpretable and need modest training data. PLSR is the standard chemometric method for soil spectra, handling many collinear

bands.

E.2 Machine learning

Random forest, Cubist, gradient boosting (XGBoost, LightGBM), support vector regression and Gaussian-process regression dominate crop classification, yield estimation and digital soil mapping. Deep learning - 1-D/2-D/3-D CNNs, LSTM/GRU recurrent networks and transformers on image time series - now gives the best crop-type and yield accuracies where large labelled datasets exist (e.g. CAP parcel declarations, Cropland Data Layer). Systematic reviews of SOC estimation report that deep models often outperform shallow networks and that the spatial distribution of samples matters more than their count.

E.3 Physically based retrieval and data assimilation

Radiative-transfer models (PROSPECT + SAIL = PROSAIL) simulate canopy reflectance from leaf chemistry and canopy structure; inverting them yields LAI, chlorophyll and water content without site-specific calibration. Crop growth models (WOFOST, DSSAT, APSIM, STICS) can be constrained by assimilating satellite LAI or soil moisture through ensemble Kalman filters or variational schemes, improving yield forecasts and making them physically consistent.

E.4 Digital soil mapping (DSM)

DSM formalises the SCORPAN model - soil as a function of soil, climate, organisms, relief, parent material, age and position - with machine learning. Covariates include satellite reflectance and indices, bare-soil composites, land-surface temperature, DEM derivatives (slope, curvature, wetness index), climate grids and geology. SoilGrids 2.0 applies quantile random forest to ~240,000 profiles and 400+ covariates to map SOC, total N, pH, CEC, bulk density and texture at six depths and 250 m with 5th/95th-percentile uncertainty maps. National products follow the same logic at finer resolution (e.g. POLARIS and gSSURGO-based maps in the USA; LUCAS-based maps in the EU).

E.5 Inferring soil biology

No satellite measures microbes. Current approaches (i) regress eDNA- or PLFA-derived indicators (alpha diversity, biomass, F:B ratio) on hyperspectral or multispectral features plus soil and climate covariates - DESIS imaging spectroscopy with Gaussian-process regression explained roughly 50% of bacterial and 40% of fungal alpha-diversity variance in forests; (ii) relate crop-health signals (NDVI adjusted for abiotic confounders using LUCAS topsoil and ERA5 climate) to fungal community composition in cereals; or (iii) use satellite-derived SOC, moisture, temperature and vegetation as habitat covariates in microbial biogeography models. Global syntheses show microbial biomass and richness-to-biomass ratios are strongly structured by soil carbon content and pH - both mappable - which is why such indirect inference has some skill. Treat outputs as hypotheses to be verified with field assays.

Part F. Evaluation and quality assurance

F.1 Validation design

- **Independent reference data**: laboratory soil analyses (LUCAS, national soil surveys, project sampling), yield monitors or harvest plots, field crop-type surveys, parcel declarations.
- **Spatial cross-validation** (blocked by field, farm or region) rather than random k-fold, to avoid optimistic bias from spatial autocorrelation; SoilGrids 2.0 explicitly adopts spatial CV.
- **Temporal transfer tests**: train on one or more seasons and test on an unseen season to mimic operational forecasting.
- **Sensor transfer tests**: check whether models calibrated on Sentinel-2 hold for Landsat or MODIS; studies find skill in calibration but limited spatial robustness across sensors.

F.2 Metrics and benchmarks

Task	Metrics	Typical published performance	Interpretation notes
Soil SOC / clay / CaCO3	R^2, RMSE, bias, RPD (SD/RMSE), RPIQ (IQR/RMSE), Lin's concordance	Satellite SOC: R^2 0.4-0.7, RPD 1.3-2.0; hyperspectral and bare-soil composites at the upper end	RPD < 1.4 poor, 1.4-2.0 fair/screening, > 2.0 good; report prediction intervals
Soil pH, CEC, total N	As above	Usually lower: R^2 0.3-0.6; CEC/pH reach 'intermediate' skill only where they covary with SOC or carbonates	Largely indirect predictions; strong local calibration needed
Crop type classification	Overall accuracy, F1 per class, kappa, confusion matrix; design-based area estimates with confidence intervals	OA 85-95% for major crops with full-season S1+S2 time series	Minor crops and early-season maps much weaker
Yield estimation	R^2, RMSE, MAPE, nRMSE vs. yield monitor / official statistics	Field-scale R^2 0.5-0.8 for cereals; county/regional R^2 often > 0.8	Check performance in anomalous (drought) years
Biophysical variables (LAI, CCC)	RMSE vs. in-situ, with uncertainty flags	LAI RMSE ~0.5-1.0	Saturation at LAI > 4-5
Soil biology indicators	Explained variance, correlation with eDNA / PLFA, cluster-level significance	Up to ~40-50% variance for alpha diversity; associations rather than predictions for F:B	Exploratory; not yet suitable for management decisions

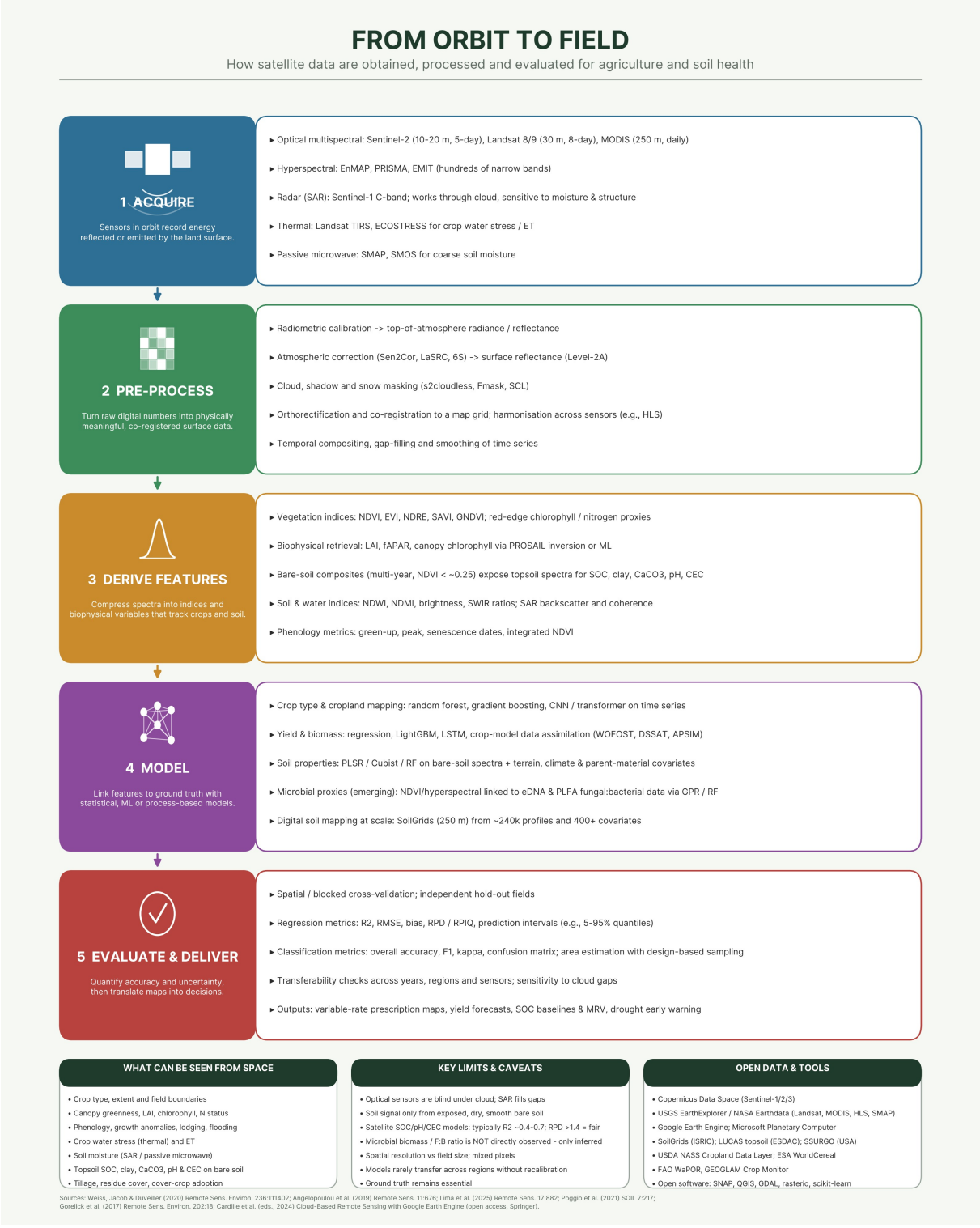
F.3 Uncertainty, documentation and delivery

Good practice includes per-pixel uncertainty (quantile regression forests, bootstrap ensembles, conformal prediction), prediction-interval coverage checks, area-of-applicability analysis for ML models, and metadata that records sensor, processing version, training data and validation design. Deliverables are typically cloud-optimised GeoTIFFs or vector management zones consumed by farm-management software, variable-rate controllers, insurers, or carbon-MRV protocols.

Part G. Worked example workflow (Sentinel-2 to topsoil SOC and pH map)

- Define area and period (e.g. 5 years of Sentinel-2 L2A over an agricultural region); load from Earth Engine or the Copernicus Data Space.
- Mask clouds/shadows with SCL + s2cloudless; compute NDVI, NBR2 and a SWIR moisture index per scene.
- Select bare, dry, residue-free pixels (NDVI < 0.25, NBR2 < 0.1, moisture index below threshold); take the per-band median to build the bare-soil composite.
- Extract composite spectra, band ratios and DEM/climate covariates at soil-sample points (e.g. LUCAS topsoil); harmonise lab methods and depths.
- Fit PLSR and random forest / Cubist models for SOC, clay, $CaCO_3$, pH and CEC; tune with spatially blocked cross-validation.
- Report R^2, RMSE, RPD/RPIQ and prediction intervals; map predictions and uncertainty; mask pixels outside the model's area of applicability.
- Validate against an independent hold-out set or a new sampling campaign; document limitations (e.g. no predictions under permanent cover).

Appendix. Process infographic



Generated by the accompanying stand-alone script *make_infographic.py*.